

CSE 331 (Spring 2026): Sample Practice Problems for Midterm 1

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1 Introduction

The following are some sample practice problems in preparation for the midterm exam 1. The students should not expect the midterm questions to cover these questions exactly. They are likely to be in a similar spirit, but not exactly the same.

Solving these questions does not guarantee that students will do well on the midterm exam.

Problem 1 (Message Authentication Code - 10 points) Suppose you are given a message authentication code scheme $\text{MAC}_B(k, m)$. This scheme, however, can only calculate the MAC of a message m only when the length of the message ℓ_m is exactly 128 bits (i.e., $|m| = \ell_m = 128$ bits). A friend of yours developed a scheme $\text{MAC}(k, M)$ that uses the MAC scheme $\text{MAC}_B(k, m)$ as a black box. His scheme can handle messages M of arbitrary length; not restricted to 128 bits only. He claims that his scheme, which functions in the following fashion, satisfies the security of a MAC scheme. **Your task is to show that this scheme is actually broken.** This is how his scheme works: Given a message M of arbitrary length, he first pads the message with some extra information called the pad to make the resulting message a multiple of 128 bits. Assume the resulting message is M_p and is of the form: $M||\text{pad}$. Also, $|M_p|$ is a multiple of 128 bits. According to his scheme, after padding M_p is decomposed into 128 bit blocks: $M_p = M_1||M_2||\dots||M_n$. Then, $\text{MAC}(K, M_p) = \text{MAC}_B(K, M_1) \oplus \text{MAC}_B(K, M_2) \oplus \dots \oplus \text{MAC}_B(K, M_n)$.

Problem 2 (Multiplicative Homomorphic Encryption - 10 points) Please describe in your own words, what is the property of a multiplicative homomorphic encryption scheme. Assume you are given the description of El Gamal Public-key cryptosystem. ***Prove that this cryptosystem is multiplicatively homomorphic.***

Problem 3 (Shared Key Between Friends - 10 points) Assume three friends Alice, Bob, and Carol want to establish a single symmetric key (K_{ABC}) among themselves. The communication channel between them is controlled by a passive adversary. ***Please describe an approach that can***

help them establish K_{ABC} only using the insecure communication channel between them.

Problem 4 (Outsourced Storage - 10 points) Assume you have a lot of readings from a sensor in your lab. The number of readings is way too many to be stored in your Chromebook, and so you decided to upload all of these readings (*readings can be assumed to be 32-bit integers*) to a cloud server, which is offering you free space. You want to run different queries on the data. However, you are worried that the cloud server in the free version may not store all your data. *Come up with a mechanism with which you can provably verify that whatever readings the server returned for your queries were actually part of the readings you had uploaded before.* For this question, you can explain your mechanism when the reading you uploaded consisted of the following: $\{1, 2, 3, 4\}$.

Problem 5 (Perfect Secrecy - 10 points) *Explain in your own words what property a cipher scheme needs to have to be called perfectly secure.*

Problem 6 (User Authentication - 10 points) We implemented a password vault that works locally. In that password vault, you used AES in the GCM mode. What is the advantage of that mode of cipher compared to AES ECB/CTR/CBC (assume you are given the diagram for those), for instance?

Problem 7 (M2M Authentication - 10 points) In class, we saw an implementation bug in an X.509 certificate chain validation implementation which perceived a certificate that expired in 1995 to be valid till 2095. Assume that the corresponding private key to the certificate was never released. *What security threat can this buggy implementation face in these modern times?*

Problem 8 (Computational Security - 10 points) In the context of a symmetric cipher, computational security is considered to relax the stringent restriction imposed by the perfect secrecy requirements. Can you give an example case in which a block cipher may be considered computationally secure but is insecure in the perfect secrecy mode?

Problem 9 (User Authentication - 10 points) For storing passwords, we learned about storing hashed passwords with salts (a fresh random number); $H(\text{password} \cdot \text{salt})$ in which $\cdot$ denotes concatenation operation. What kind of attacks can an adversary launch on a stolen password file if the passwords are not salted before hashing?

Problem 10 (Random one-time Pad - 10 points) In random one-time pad, for encrypting a message m of length ℓ , we randomly generate a fresh value v of length ℓ . We then encrypt the message m as follows $m \oplus v$ where $\oplus$

denotes the XOR operation. What if we use the same value v for encrypting two different messages m_1 and m_2 of equal lengths ℓ ? Are there any security implications of this?

Problem 11 (Cipher Security - 10 points) Consider you're an adversary who has been listening in on a conversation between two friends. You somehow figured out that in the protocol they are using the first encrypted message they sent is always the encryption of the word "HELLO". When designing a cipher to be used in this protocol, what type of attacker should the designer be protecting against (e.g., chosen ciphertext attack)?

Problem 12 (Digital Signature - 10 points) Why does one sign the hash of a message instead of the message itself? Explain why signing the hash of a message is more beneficial than the message itself. Also, explain what property of a hash function (i.e., one-wayness, weak collision resistance, strong collision resistance) is expected when using it in the context of a digital signature.

Problem 13 (Non-repudiation - 10 points) Non-repudiation is a property where the signing party (or, tag-generating party) cannot refuse that they signed a message. We discussed that a digital signature scheme can provide non-repudiation whereas a message authentication code (MAC) in the symmetric setting cannot. Explain why.

Problem 14 (RSA and IND-CPA - 10 points) Show that the original RSA construction does not have IND-CPA Security.

Problem 15 (User Authentication - 10 points) Assume you're given the design of the Honeyword scheme. What is the overarching goal of this scheme? Is it to protect against offline password attacks? Explain what is offline and online password attacks.